

Customer Churn Analysis & Retention Insights

A Data Analytics Case Study for a Subscription-Based OTT Platform

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Executive Summary

This report presents an end-to-end churn analysis built on a relational customer dataset covering subscriber demographics, subscription details, and support interactions. The objective was to identify which customer segments are most likely to churn, understand the behavioral and operational drivers behind cancellations, and translate the findings into concrete retention actions. The analysis pipeline was built using Python (pandas, NumPy, sqlite3) for data extraction and cleaning, SQL for relational querying, and Matplotlib/Seaborn for exploratory visualization.

Headline Metrics

28.6% Overall Churn Rate	71.4% Retention Rate	1,451 days Avg. Customer Tenure	18% Revenue Lost to Churn
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Business Problem

Subscription businesses live or die by retention — acquiring a new customer typically costs far more than keeping an existing one. The goal of this project was to act as a data analyst would in a real OTT/streaming business: pull data from a relational database (customer, subscription, and support tables), engineer churn-relevant features, and surface which customers are at risk before they cancel.

Data & Methodology

- Data extraction: connected to a SQLite database and pulled three relational tables — Customer, Subscription, and Support — using Python and SQL joins.
- Data cleaning: resolved missing values (e.g., imputing country from existing state data), corrected data types, removed duplicate records, and validated key fields.
- Feature engineering: derived a churn flag from cancellation dates, calculated customer tenure, monthly recurring revenue at risk, and composite churn-risk categories (High/Medium/Low).
- Exploratory analysis: used group-by aggregation and pivot tables to break down churn by plan type, contract type, geography, and support-escalation history.
- Visualization: built time-trend and correlation charts in Matplotlib/Seaborn to communicate patterns clearly.

Database Schema

The source data lives in a relational SQLite database (customer_churn) split across three linked tables, joined on customerid:

Customer	Subscription	Support
customerid name country	customerid subscription_start_date subscription_type	customerid complaint_date escalations

Customer	Subscription	Support
state gender date of birth interests pincode	renewal_date plan_type contract_type cancellation_date cancellation_reason monthly_charges cltv churn_score	csat_score comment

Calculated Metrics

Each KPI below was derived using SQL aggregation and pandas group-by logic on the joined tables:

KPI	Formula
Churn rate	churned customers ÷ total customers
Churn by plan type	churn rate grouped by plan_type (Basic / Standard / Premium)
Churn by state	churn rate grouped by country, state
Retention rate	1 – churn rate
ARPU	SUM(monthly_charges) ÷ COUNT(active customerid)
Average customer tenure	AVG(DATEDIFF(cancellation_date OR today, subscription_start_date))
Revenue at risk	SUM(monthly_charges) WHERE churn_score > 70
Escalation rate	SUM(escalations) ÷ COUNT(complaints) × 100
Avg. complaints per customer	COUNT(complaints) ÷ COUNT(DISTINCT customerid)
Escalations → churn correlation	churn rate where escalations ≥ 1 vs. escalations = 0

Key Findings

Metric	Value	Insight
Overall churn rate	28.6%	Roughly 3 in 10 subscribers churned; retention sits at 71.4%.
Monthly vs. annual churn	55.6% vs 8.3%	Monthly-contract subscribers churn at nearly 7x the rate of annual-contract subscribers — the single strongest churn driver found.
Plan type	Basic plan highest	Basic-tier users show the most cancellations, but since ARPU on this tier is low, the revenue impact is smaller than the churn count suggests.
Geography	Karnataka highest	Karnataka recorded the largest concentration of churned customers, with a cancellation spike in September — worth investigating for pricing or service-quality causes.

Metric	Value	Insight
Support escalations	Correlated with churn	Customers with one or more escalated support tickets churn at a visibly higher rate than those with none, pointing to unresolved service issues as a churn trigger.
Revenue impact	18% of revenue at risk	Estimated CLTV erosion of ~2,047 units and recurring revenue leakage tied directly to the identified at-risk cohort.

Recommendations

- Prioritize a contract-migration campaign — incentivize monthly subscribers to switch to annual plans, since this segment shows the largest churn gap (55.6% vs 8.3%).
- Investigate the Karnataka / September churn spike specifically: check for regional pricing changes, service outages, or a spike in complaints during that period.
- Route customers with escalated support tickets into a proactive retention workflow (priority callback, discount, or issue resolution SLA) before they reach cancellation.
- Build a simple churn-risk scoring view (High/Medium/Low) so the retention/CX team can proactively target the highest-CLTV at-risk customers rather than reacting after cancellation.
- Track competitor movement where cancellation reasons cite switching, to understand feature or pricing gaps.

Next Steps

The natural extension of this analysis is a predictive churn model (e.g., logistic regression or gradient boosting) trained on the engineered features, to score every active customer with a churn probability rather than relying on retrospective segmentation. A lightweight interactive dashboard (Streamlit) is also planned to let a non-technical stakeholder filter churn by plan, region, and support history in real time.

Tools & Stack

Python (pandas, NumPy, sqlite3) · SQL · Matplotlib · Seaborn · Jupyter Notebook